Supplementary Information

Combining detectors that fail differently protects screens for machine-written scholarly text against adaptive authors · Ook Lee

Contents: 1 Corpus · 2 Full AUC tables · 3 Bootstrap intervals · 4 Operating points · 5 Judge · 6 Economics papers · 7 Guided attacks · 8 Count screen transfer · 9 Prompts and procedures. *Proposed defence components. Abbreviations: RoBRT = RoBERTa classifier; Count = count screen (clustering features); LogLik = one-sided log-likelihood; Binoc. = Binoculars; FDGPT = Fast-DetectGPT; 2-sided = two-sided log-likelihood; 2-s. ent. = two-sided entropy; Mean3 = mean of standardized FDGPT, RoBERTa and Count; Mean4 = Mean3 plus 2-sided; Top2 = mean of the two largest of the Mean4 components.

1. Corpus

Supplementary Table 1 | Documents. Word counts before truncation to 2,000 tokens. DeepSeek papers: 200 generated, of which 100 with at least 2,400 words were attacked.

Source	Documents	Median words
Human, medicine and computer science (PMC, arXiv cs)	500 collected / 474 analysed	4,299
GPT-4o	245	3,639
Claude Sonnet 5.5	151	3,476
Llama 3.3 70B Instruct	202	3,002
DeepSeek V4.1 Flash	190	2,641
Human, economics (arXiv econ.*)	130	10,772
GPT-4o, economics topics	100	3,809
Claude, economics topics	100	3,605
Llama, economics topics	100	3,082
DeepSeek, economics topics	94	2,566

2. Full AUC tables (medicine and computer science)

Negatives: 186 evaluation humans. Positives: 100 attacked papers per cell (guided attacks: 100 per generator for the three original generators). Training for each target excluded the target generator, Llama and DeepSeek.

Supplementary Table 2 | Claude papers, AUC.

Attack	Count	LogLik	Binoc.	FDGPT	TF-IDF	RoBRT	2-sided*	2-s. ent.	Mean3	Mean4*	Top2*
none	0.76	0.56	0.67	0.94	0.92	0.94	0.46	0.51	0.94	0.93	0.90
edit e=0.25	0.69	0.52	0.67	0.92	0.92	0.94	0.48	0.54	0.92	0.91	0.88
edit e=0.5	0.65	0.49	0.66	0.91	0.92	0.92	0.51	0.57	0.89	0.89	0.84
edit e=0.75	0.63	0.47	0.65	0.89	0.92	0.92	0.52	0.59	0.88	0.88	0.84
edit e=1	0.59	0.46	0.65	0.90	0.92	0.90	0.53	0.59	0.86	0.86	0.82
LLM rewrite	0.80	0.65	0.53	0.95	1.00	0.97	0.47	0.48	0.97	0.96	0.94
rewrite+edit 0.5	0.67	0.57	0.54	0.92	1.00	0.96	0.47	0.52	0.93	0.92	0.90
rewrite+edit 1	0.67	0.55	0.55	0.93	1.00	0.94	0.47	0.54	0.92	0.91	0.88
rewrite twice	0.83	0.62	0.46	0.97	1.00	0.99	0.46	0.50	0.98	0.98	0.96
3% typos	0.75	0.43	0.62	0.83	0.92	0.92	0.53	0.57	0.91	0.91	0.88
10% typos	0.74	0.24	0.65	0.77	0.92	0.86	0.69	0.75	0.87	0.90	0.86
30% human sent.	0.63	0.14	0.43	0.69	0.78	0.79	0.80	0.84	0.76	0.86	0.81
50% human	0.56	0.05	0.35	0.61	0.62	0.53	0.93	0.95	0.56	0.82	0.81

sent.											
guided rewrite	0.83	0.63	0.47	0.93	1.00	0.98	0.45	0.46	0.97	0.97	0.95
guided mixing	0.59	0.03	0.28	0.35	0.59	0.58	0.96	0.95	0.49	0.79	0.81

Supplementary Table 3 | GPT-4o papers, AUC.

Attack	Count	LogLik	Binoc.	FDGPT	TF-IDF	RoBRT	2-sided*	2-s. ent.	Mean3	Mean4*	Top2*
none	0.86	0.93	0.83	1.00	0.99	0.99	0.78	0.46	1.00	1.00	1.00
edit e=0.25	0.80	0.91	0.83	1.00	0.99	0.99	0.72	0.45	1.00	1.00	0.99
edit e=0.5	0.76	0.90	0.83	1.00	0.98	0.98	0.69	0.44	1.00	1.00	0.99
edit e=0.75	0.72	0.89	0.83	1.00	0.98	0.98	0.67	0.43	1.00	1.00	0.99
edit e=1	0.70	0.88	0.83	1.00	0.98	0.98	0.64	0.43	1.00	1.00	0.99
LLM rewrite	0.86	0.94	0.81	1.00	0.85	0.97	0.79	0.48	1.00	1.00	1.00
rewrite+edit 0.5	0.73	0.89	0.81	1.00	0.87	0.95	0.69	0.44	1.00	1.00	0.99
rewrite+edit 1	0.67	0.88	0.80	1.00	0.87	0.93	0.65	0.43	1.00	0.99	0.98
rewrite twice	0.83	0.89	0.75	1.00	0.75	0.97	0.68	0.45	1.00	1.00	1.00
3% typos	0.86	0.85	0.82	1.00	0.98	0.99	0.59	0.45	1.00	1.00	0.99
10% typos	0.85	0.62	0.82	1.00	0.98	0.97	0.34	0.53	0.99	0.99	0.97
30% human sent.	0.77	0.37	0.61	0.97	0.88	0.94	0.54	0.68	0.97	0.96	0.92
50% human sent.	0.66	0.12	0.52	0.92	0.74	0.84	0.83	0.92	0.90	0.95	0.88
guided rewrite	0.88	0.89	0.73	1.00	0.89	0.97	0.69	0.46	1.00	1.00	1.00
guided mixing	0.64	0.08	0.39	0.80	0.75	0.85	0.89	0.93	0.81	0.93	0.86

Supplementary Table 4 | Llama papers, AUC.

Attack	Count	LogLik	Binoc.	FDGPT	TF-IDF	RoBRT	2-sided*	2-s. ent.	Mean3	Mean4*	Top2*
none	0.90	0.99	0.97	1.00	1.00	0.96	0.97	0.79	1.00	1.00	1.00
edit e=0.25	0.83	0.99	0.96	1.00	1.00	0.96	0.95	0.72	1.00	1.00	1.00
edit e=0.5	0.78	0.98	0.96	1.00	1.00	0.95	0.94	0.68	1.00	1.00	1.00
edit e=0.75	0.77	0.98	0.96	1.00	1.00	0.94	0.93	0.66	1.00	1.00	1.00
edit e=1	0.72	0.98	0.95	1.00	1.00	0.94	0.92	0.64	1.00	1.00	1.00
LLM rewrite	0.88	0.99	0.96	1.00	1.00	0.97	0.97	0.74	1.00	1.00	1.00
rewrite+edit 0.5	0.78	0.98	0.96	1.00	1.00	0.96	0.93	0.64	1.00	1.00	1.00
rewrite+edit 1	0.73	0.98	0.95	1.00	1.00	0.94	0.91	0.61	1.00	1.00	0.99
rewrite twice	0.89	0.97	0.92	1.00	1.00	0.99	0.90	0.53	1.00	1.00	1.00
3% typos	0.90	0.98	0.96	1.00	1.00	0.96	0.92	0.66	1.00	1.00	0.99
10% typos	0.89	0.92	0.95	1.00	0.99	0.95	0.75	0.43	1.00	1.00	0.99
30% human sent.	0.78	0.57	0.73	1.00	0.93	0.88	0.44	0.59	0.98	0.97	0.96
50% human sent.	0.64	0.23	0.61	0.98	0.80	0.76	0.66	0.88	0.91	0.91	0.88
guided rewrite	0.91	0.98	0.93	1.00	1.00	0.99	0.92	0.66	1.00	1.00	1.00
guided mixing	0.64	0.17	0.47	0.95	0.79	0.76	0.76	0.88	0.87	0.89	0.85

Supplementary Table 5 | DeepSeek papers, AUC.

Attack	Count	LogLik	Binoc.	FDGPT	TF-IDF	RoBRT	2-sided*	2-s. ent.	Mean3	Mean4*	Top2*
none	0.85	0.77	0.75	0.99	1.00	0.99	0.54	0.47	1.00	1.00	1.00
edit e=0.25	0.82	0.73	0.74	0.99	1.00	0.99	0.51	0.48	1.00	1.00	1.00
edit e=0.5	0.80	0.70	0.74	0.99	1.00	0.99	0.49	0.48	1.00	1.00	1.00
edit e=0.75	0.76	0.69	0.74	0.99	1.00	0.99	0.48	0.49	1.00	1.00	1.00
edit e=1	0.72	0.68	0.74	0.99	1.00	0.99	0.48	0.49	1.00	1.00	1.00
LLM rewrite	0.84	0.81	0.67	0.99	1.00	0.99	0.59	0.45	1.00	1.00	1.00
rewrite+edit 0.5	0.75	0.74	0.65	0.99	1.00	0.99	0.50	0.47	1.00	1.00	1.00
rewrite+edit 1	0.72	0.71	0.64	0.98	1.00	0.99	0.49	0.47	1.00	1.00	1.00
rewrite twice	0.86	0.76	0.60	1.00	1.00	1.00	0.53	0.46	1.00	1.00	1.00
3% typos	0.85	0.64	0.71	0.96	1.00	0.99	0.47	0.50	1.00	1.00	1.00
10% typos	0.85	0.42	0.75	0.94	1.00	0.99	0.50	0.65	1.00	1.00	1.00
30% human sent.	0.74	0.26	0.55	0.87	0.99	0.97	0.65	0.77	0.98	0.99	0.98
50% human sent.	0.68	0.08	0.45	0.76	0.89	0.87	0.89	0.93	0.89	0.95	0.93

3. Bootstrap intervals for key cells

Supplementary Table 6 | AUC with 95% bootstrap intervals (400 resamples of documents within each class).

Generator	Attack	FDGPT	RoBRT	Count	2-sided*	Mean4*	Top2*
Claude	none	0.94 [0.91, 0.96]	0.94 [0.92, 0.97]	0.76 [0.70, 0.81]	0.46 [0.39, 0.53]	0.93 [0.91, 0.96]	0.90 [0.87, 0.94]
Claude	50% human sent.	0.61 [0.54, 0.67]	0.53 [0.45, 0.59]	0.56 [0.50, 0.63]	0.93 [0.90, 0.96]	0.82 [0.76, 0.86]	0.81 [0.75, 0.85]
Claude	guided mixing	0.35 [0.28, 0.41]	0.58 [0.51, 0.64]	0.59 [0.52, 0.65]	0.96 [0.93, 0.98]	0.79 [0.74, 0.84]	0.81 [0.76, 0.85]
Claude	10% typos	0.77 [0.72, 0.83]	0.86 [0.81, 0.90]	0.74 [0.67, 0.79]	0.69 [0.62, 0.76]	0.90 [0.86, 0.93]	0.86 [0.82, 0.90]
GPT-4o	none	1.00 [1.00, 1.00]	0.99 [0.98, 0.99]	0.86 [0.82, 0.90]	0.78 [0.73, 0.82]	1.00 [1.00, 1.00]	1.00 [0.99, 1.00]
GPT-4o	50% human sent.	0.92 [0.88, 0.95]	0.84 [0.79, 0.89]	0.66 [0.60, 0.72]	0.83 [0.78, 0.87]	0.95 [0.93, 0.97]	0.88 [0.84, 0.92]
GPT-4o	guided mixing	0.80 [0.75, 0.85]	0.85 [0.80, 0.89]	0.64 [0.57, 0.70]	0.89 [0.86, 0.93]	0.93 [0.90, 0.96]	0.86 [0.82, 0.90]
GPT-4o	10% typos	1.00 [1.00, 1.00]	0.97 [0.95, 0.99]	0.85 [0.80, 0.89]	0.34 [0.28, 0.40]	0.99 [0.98, 0.99]	0.97 [0.95, 0.98]
Llama	none	1.00 [1.00, 1.00]	0.96 [0.94, 0.98]	0.90 [0.86, 0.93]	0.97 [0.95, 0.99]	1.00 [1.00, 1.00]	1.00 [1.00, 1.00]
Llama	50% human sent.	0.98 [0.97, 0.99]	0.76 [0.71, 0.82]	0.64 [0.57, 0.71]	0.66 [0.60, 0.73]	0.91 [0.88, 0.94]	0.88 [0.85, 0.91]
Llama	guided mixing	0.95 [0.92, 0.97]	0.76 [0.69, 0.81]	0.64 [0.57, 0.70]	0.76 [0.71, 0.82]	0.89 [0.85, 0.93]	0.85 [0.80, 0.89]
Llama	10% typos	1.00 [1.00, 1.00]	0.95 [0.93, 0.97]	0.89 [0.85, 0.93]	0.75 [0.69, 0.81]	1.00 [1.00, 1.00]	0.99 [0.98, 1.00]
DeepSeek	none	0.99 [0.99, 1.00]	0.99 [0.98, 1.00]	0.85 [0.81, 0.90]	0.54 [0.47, 0.61]	1.00 [1.00, 1.00]	1.00 [1.00, 1.00]
DeepSeek	50% human sent.	0.76 [0.69, 0.82]	0.87 [0.82, 0.90]	0.68 [0.62, 0.75]	0.89 [0.85, 0.92]	0.95 [0.93, 0.97]	0.93 [0.90, 0.96]
DeepSeek	10% typos	0.94 [0.91, 0.97]	0.99 [0.97, 1.00]	0.85 [0.80, 0.89]	0.50 [0.42, 0.56]	1.00 [0.99, 1.00]	1.00 [0.99, 1.00]

4. Operating points

Supplementary Table 7 | Share of machine papers flagged at the 95th percentile of 100 calibration humans; last row: false-positive rate on the evaluation humans.

Generator	Attack	FDGPT	RoBRT	TF-IDF	Count	2-sided*	Mean4*	Top2*
Claude	none	0.69	0.82	0.60	0.18	0.04	0.57	0.41
Claude	edit e=1	0.54	0.61	0.60	0.03	0.09	0.20	0.09
Claude	rewrite twice	0.87	0.98	1.00	0.23	0.03	0.82	0.61

Claude	50% human sent.	0.12	0.05	0.01	0.00	0.79	0.17	0.10
Claude	guided mixing	0.00	0.08	0.04	0.01	0.85	0.09	0.09
GPT-4o	none	1.00	0.92	0.95	0.30	0.25	1.00	1.00
GPT-4o	edit e=1	1.00	0.89	0.93	0.03	0.18	0.99	0.97
GPT-4o	rewrite twice	1.00	0.83	0.25	0.22	0.16	1.00	1.00
GPT-4o	50% human sent.	0.70	0.34	0.24	0.03	0.40	0.81	0.28
GPT-4o	guided mixing	0.25	0.31	0.34	0.04	0.57	0.67	0.31
Llama	none	1.00	0.87	0.96	0.41	0.92	1.00	1.00
Llama	edit e=1	1.00	0.80	0.95	0.06	0.66	1.00	0.98
Llama	rewrite twice	1.00	0.99	1.00	0.41	0.61	1.00	0.99
Llama	50% human sent.	0.92	0.21	0.17	0.02	0.23	0.46	0.42
Llama	guided mixing	0.71	0.25	0.15	0.01	0.43	0.40	0.33
DeepSeek	none	0.95	0.99	1.00	0.26	0.08	1.00	1.00
DeepSeek	edit e=1	0.89	0.98	1.00	0.06	0.06	1.00	0.99
DeepSeek	rewrite twice	0.98	1.00	1.00	0.18	0.06	1.00	1.00
DeepSeek	50% human sent.	0.39	0.48	0.25	0.00	0.62	0.74	0.62
FPR	evaluation humans (mean over targets)	0.030	0.059	0.041	0.032	0.088	0.044	0.043

5. Language-model judge

Supplementary Table 8 | Gemini 2.5 Pro judge. Negatives: 200 evaluation humans. Runs on mixing were limited by a spending cap.

Generator	Attack	n	AUC	Share ≥ 50
Claude	edit e=1	100	0.83	0.33
Claude	rewrite+edit 1	99	0.96	0.83
Claude	none	100	0.82	0.15
Claude	LLM rewrite	99	0.97	0.76
GPT-4o	edit e=1	100	0.98	0.80
GPT-4o	rewrite+edit 1	100	1.00	0.99
GPT-4o	none	100	0.95	0.71
GPT-4o	LLM rewrite	100	1.00	0.97
Llama	edit e=1	100	0.98	0.87
Llama	rewrite+edit 1	99	1.00	1.00
Llama	none	100	0.98	0.94
Llama	LLM rewrite	99	1.00	1.00
Claude	50% human sent.	99	0.96	0.85
GPT-4o	50% human sent.	100	0.96	0.81
Llama	50% human sent.	64	1.00	0.94
Claude	guided mixing	100	0.96	0.80
GPT-4o	guided mixing	100	0.98	0.91
Llama	guided mixing	100	0.99	0.91
DeepSeek	none	100	0.91	0.48
DeepSeek	LLM rewrite	100	0.97	0.90
DeepSeek	50% human sent.	100	0.97	0.82
Human	none	200	–	0.01

6. Economics papers

Supplementary Table 9 | Economics papers, AUC. Negatives: 130 economics human papers; calibration: 86 calibration humans from the main fields. Supervised models trained on medicine and computer science (human, GPT-4o, Claude). Human economics papers were all converted from PDF, so supervised scores here are likely inflated by structural artifacts (main text).

Generator	Attack	FDGPT	RoBRT	TF-IDF	Count	2-sided*	Mean4*	Top2*
Claude	none	0.93	1.00	1.00	0.76	0.54	1.00	1.00
Claude	50% human sent.	0.49	0.95	0.97	0.59	0.99	0.98	0.98
Claude	10% typos	0.84	1.00	1.00	0.76	0.90	1.00	1.00
GPT-4o	none	1.00	1.00	1.00	0.93	0.43	1.00	1.00
GPT-4o	50% human sent.	0.91	0.99	1.00	0.72	0.94	1.00	0.99
GPT-4o	10% typos	1.00	1.00	1.00	0.93	0.36	1.00	1.00
Llama	none	1.00	1.00	1.00	0.81	0.92	1.00	1.00
Llama	50% human sent.	0.97	0.90	0.92	0.65	0.80	0.97	0.96
Llama	10% typos	1.00	0.99	1.00	0.81	0.48	1.00	1.00
DeepSeek	none	0.97	1.00	1.00	0.82	0.32	1.00	1.00
DeepSeek	50% human sent.	0.61	0.98	0.98	0.62	0.98	0.99	0.99
DeepSeek	10% typos	0.92	1.00	1.00	0.81	0.76	1.00	1.00

7. Guided attacks

Supplementary Table 10 | Mean Fast-DetectGPT score (higher = more machine-like; human papers average about 0.06). Selection lowers the score far more for mixing than for rewriting.

Generator	Original	Rewrite (one)	Guided rewrite (best of 4)	Mixing (one)	Guided mixing (best of 4)
Claude	2.26	2.52	2.22	0.45	-0.47
GPT-4o	6.39	6.36	5.82	2.29	1.19
Llama	7.95	8.41	7.64	3.50	2.26

8. Count screen: transfer between generators (full corpus, earlier protocol)

Supplementary Table 11 | Count screen, AUC (95% bootstrap interval) with five-fold cross-validation grouped by source paper, before the calibration split was introduced.

Train → test	Rates	Dispersion	Both
gpt4o → gpt4o	0.864 [0.84, 0.89]	0.870 [0.84, 0.90]	0.945 [0.93, 0.96]
gpt4o → claude	0.745 [0.70, 0.79]	0.702 [0.65, 0.74]	0.771 [0.73, 0.81]
gpt4o → llama	0.680 [0.64, 0.72]	0.656 [0.61, 0.70]	0.711 [0.67, 0.75]
claude → gpt4o	0.769 [0.73, 0.80]	0.796 [0.76, 0.83]	0.861 [0.83, 0.89]
claude → claude	0.854 [0.81, 0.89]	0.684 [0.63, 0.73]	0.866 [0.83, 0.90]
claude → llama	0.775 [0.74, 0.81]	0.676 [0.64, 0.72]	0.824 [0.79, 0.86]
llama → gpt4o	0.752 [0.72, 0.78]	0.695 [0.65, 0.74]	0.830 [0.80, 0.86]
llama → claude	0.744 [0.70, 0.79]	0.591 [0.54, 0.64]	0.770 [0.73, 0.81]
llama → llama	0.909 [0.88, 0.93]	0.763 [0.72, 0.80]	0.935 [0.91, 0.96]

Supplementary Table 12 | Count screen on an unseen generator.

Target	Baseline	Clustering	Difference
GPT-4o	0.853 [0.82, 0.88]	0.886 [0.86, 0.91]	+0.032 [+0.015, +0.052]
Claude	0.781 [0.74, 0.82]	0.810 [0.76, 0.85]	+0.029 [-0.003, +0.056]
Llama	0.792 [0.75, 0.83]	0.861 [0.84, 0.89]	+0.069 [+0.045, +0.093]

9. Prompts and procedures

Rewrite (attacks 'LLM rewrite', 'rewrite + edit', guided rewrite with even k). Rewrite the following passage from a scientific paper so that it reads as if it had been written by an experienced human researcher. Keep the meaning, the technical content and the citation markers. The rewritten passage should be about N words long. Output only the rewritten passage. [passage]
Second rewrite (attack 'rewrite twice', guided rewrite with odd k; temperature 1). Rewrite the following passage so that it uses clearly different vocabulary and sentence structure from the original, while keeping the meaning, the technical content and the citation markers. The result should be about N words long. Output only the rewritten passage. [passage]
Judge (Gemini 2.5 Pro, temperature 0). Below is an excerpt from a scientific paper. Estimate the probability that it was written by an AI language model (for example ChatGPT, Claude or Llama) rather than by a human author. Answer with a single number from 0 to 100, where 0 means certainly human-written and 100 means certainly AI-written. Output only the number. Excerpt: [excerpt]
Procedure. Rewrites were done in chunks of 800 tokens by Llama 3.3 70B Instruct. Mixing replaces sentences with sentences of similar length from human papers of the same field that are neither evaluation humans nor the paper's own source. Zero-shot scores used Falcon-7B (observer) and Falcon-7B-instruct (performer) on the first 1,024 tokens. RoBERTa: roberta-base, two epochs, learning rate 2×10^{-5}, batch 16, maximum length 512. Compute: about US$118 of API calls in total for generation, attacks and judging, and about US$4 of GPU time for the final scoring run. All code and derived data: https://doi.org/10.5281/zenodo.23664508.